

On Weakly Monotone Decreasing Super-Solutions to a Coupled Hardy-Sobolev Integral System

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Abstract In this article we consider weakly monotone decreasing super-solutions to a weighted integral system of the Hardy-Littlewood-Sobolev type:

$$\begin{cases} u(x) \geq \int_{\mathbb{R}^n} \frac{\phi(\xi)v(\xi)^q u(\xi)^r + \Gamma_1(\xi, u, v)}{|x-\xi|^{n-\alpha} |\xi|^{\sigma_1}} d\xi, & \text{in } \mathbb{R}^n, \\ v(x) \geq \int_{\mathbb{R}^n} \frac{\psi(\xi)u(\xi)^p v(\xi)^s + \Gamma_2(\xi, u, v)}{|x-\xi|^{n-\alpha} |\xi|^{\sigma_2}} d\xi, & \text{in } \mathbb{R}^n \end{cases}$$

where Γ_1, Γ_2, ϕ and ψ are non-negative functions satisfying

$$\liminf_{|x| \rightarrow \infty} \phi(x) > 0 \quad \text{and} \quad \liminf_{|x| \rightarrow \infty} \psi(x) > 0$$

in $n \geq 3$ dimensions with $0 < \alpha < n$, $\sigma_1, \sigma_2 \in [0, \alpha)$, $p, q, r, s \geq 0$ and $pq > (1-s)(1-r)$. It will be shown that the system above admits no (weakly) monotone decreasing solutions whenever $pq \leq (1-s)(1-r)$ and $r, s \in [0, 1]$. We shall give a priori decay estimates on solutions of the aforementioned class and show that they vanish at infinity no slower, and no quicker, than inverse powers of $|x|$. If these solutions u and v are fully non-integrable, we also prove that these estimates are optimal. In particular, we show that such decay solutions to the above system have the slow-rates. Hence, we shall extend the asymptotic results for decay solutions from [1] to our more general system. In simple cases, we give sufficient conditions for the non-existence of decay/integrable solutions to the above system.

Keywords: Hardy-Sobolev integral systems, mixed power systems, poly-harmonic systems, weakly monotone decreasing solutions, non-existence, Bose-Einstein condensates, nonlinear analysis.

Mathematics Subject Classification: Primary: 35B40, 35B53, 45G15, 45M05; Secondary: 35J91.

1 Introduction

In this article we derive decay estimates and non-existence criteria regarding single signed (positive) super-solutions to a certain Hardy-Sobolev type integral system involving Riesz potentials. Our primary concern will be generalizing a series of beautiful estimates established by J. Villavert in [1]. The key observation (for most of our results) will be to allow a type of “weak monotonicity”, generalizing the notion of a decay solution, similar to Harnack’s inequality for domains of infinite Lebesgue measure in which we allow the constant to depend on the function, thereby weakening some of the assumptions made in [1]. More precisely, the aim is to study the decay rates of positive measurable functions on $\mathbb{R}^n$ that satisfy the generalized integral system

$$\begin{cases} u(x) \geq \int_{\mathbb{R}^n} \frac{\phi(\xi)v(\xi)^q u(\xi)^r + \Gamma_1(\xi, u, v)}{|x - \xi|^{n-\alpha} |\xi|^{\sigma_1}} d\xi, & \text{in } \mathbb{R}^n, \\ v(x) \geq \int_{\mathbb{R}^n} \frac{\psi(\xi)u(\xi)^p v(\xi)^s + \Gamma_2(\xi, u, v)}{|x - \xi|^{n-\alpha} |\xi|^{\sigma_2}} d\xi & \text{in } \mathbb{R}^n \end{cases} \quad (1.1)$$

where Γ_1, Γ_2, ϕ and ψ are non-negative functions satisfying

$$\liminf_{|x| \rightarrow \infty} \phi(x) > 0 \quad \text{and} \quad \liminf_{|x| \rightarrow \infty} \psi(x) > 0. \quad (1.2)$$

We study system (1.1) when $n \geq 3$ is an integer, $\alpha \in (0, n)$, $0 \leq \sigma_1, \sigma_2 < \alpha$, and $p, q, r, s \geq 0$. We also impose the condition

$$\begin{cases} pq > (1-s)(1-r), & \text{if } r, s \in [0, 1], \\ pq \geq 1, & \text{otherwise.} \end{cases} \quad (1.3)$$

Our system is comparable to those considered in [1], [3], [5], [14] and [15]. It is worth noting that (1.1) is related to a fourth order equation studied as a special case in [12]. In particular, we observe that by taking $r = s = 0$, $\phi \equiv \psi \equiv 1$, and $\Gamma_{1,2} \equiv 0$ we recover the integral/differential system from [1]. In much of the literature, it is assumed furthermore that the system is *unweighted* (see [5]), i.e. $\sigma_1 = \sigma_2 = 0$. It is our intention to generalize the estimates established by Villavert in [1] and Lei-Li in [5], as well as the non-existence results from [11]. As pointed out by the author of [3], the system above arises often in the study of Bose-Einstein condensates and is a generalization of a differential system found frequently in nonlinear optics. In much of the physical applications, the integral system (1.1) occurs with power nonlinearities, i.e. Γ_1 and Γ_2 have the

form

$$\Gamma_1 = \sum_{j=1}^{n_1} v^{q_j} u^{r_j} \quad \text{and} \quad \Gamma_2 = \sum_{j=1}^{n_2} u^{p_j} v^{s_j}.$$

where $q_j, r_j, s_j, p_j \in \mathbb{R}$.

It was proven by Villavert in [11, THM-1] (see [1] for a remark on this) that system (1.1) is closely related (and often equivalent, under appropriate regularity conditions) to the fractional-differential system

$$\begin{cases} (-\Delta)^{\alpha/2} u(x) \equiv \frac{\phi(x)v(x)^q u(x)^r + \Gamma_1(x, u, v)}{|x|^{\sigma_1}} & \text{in } \mathbb{R}^n \setminus \{0\} \\ (-\Delta)^{\alpha/2} v(x) \equiv \frac{\psi(x)u(x)^p v(x)^s + \Gamma_2(x, u, v)}{|x|^{\sigma_2}} & \text{in } \mathbb{R}^n \setminus \{0\} \end{cases} \quad (1.4)$$

and thus, by studying solutions to the aforementioned integral system, we are also analyzing functions that solve certain elliptic systems in $\mathbb{R}^n$.

As mentioned previously, we are mostly concerned with positive solutions satisfying a certain “weakly monotone” property, that we define below. Our system is well understood when $r = s = 0$, $\phi, \psi \equiv 1$ and $\Gamma_{1,2} \equiv 0$, although most results are only known to be applicable when the solutions u and v are positive, bounded and decaying¹. In lieu of these assumptions, we introduce *weakly monotone decreasing* functions.

Definition. A function $f : \mathbb{R}^n \rightarrow (0, \infty]$ is called *weakly monotone decreasing* if there exists $C > 0$ and $R > 0$ such that $f(x) \leq C f(y)$ whenever $|x| \geq |y| \geq R$.

The set of all weakly monotone decreasing functions on $\mathbb{R}^n$ shall henceforth be denoted by $W_m(\mathbb{R}^n)$. In [1] and [5] a function $f : \mathbb{R}^n \rightarrow (0, \infty)$ is said to be a decay function (or solution) if $f(x) \simeq |x|^{-\theta}$ for some $\theta > 0$. Recall that we use the notation $f(x) \simeq g(x)$ if two functions $f, g : \mathbb{R}^n \rightarrow \mathbb{C}$ satisfy

$$c|g(x)| \leq |f(x)| \leq C|g(x)|, \quad \text{as } |x| \rightarrow \infty$$

for some constants $c, C > 0$. Similarly, we shall write $f(x) \lesssim g(x)$ to say that there exists a constant $C > 0$ such that

$$|f(x)| \leq C|g(x)|, \quad \text{as } |x| \rightarrow \infty$$

and likewise for $f(x) \gtrsim g(x)$.

1. We define this below, alternatively one can see [1, §1].

Remark 1. As stated in the first paragraph, weakly monotone decreasing functions generalize the notion of a decay function. Certainly, if $f(x) \simeq |x|^{-\theta}$ for some $\theta > 0$ we may take $A, B, R > 0$ such that

$$A|x|^{-\theta} \leq f(x) \leq B|x|^{-\theta}, \quad \text{whenever } |x| \geq R.$$

Hence, if $R \leq |y| \leq |x|$, we have

$$f(x) \leq B|x|^{-\theta} \leq B|y|^{-\theta} = \frac{BA}{A}|y|^{-\theta} \leq \frac{B}{A}f(y)$$

from which we obtain $f \in \mathbb{W}_m(\mathbb{R}^n)$.

The constants found in system (1.1) also give rise to two real numbers that play crucial roles in our asymptotic analysis of these solution pairs (u, v) . Put

$$r_0 := \begin{cases} \frac{p(\alpha - \sigma_1) + (\alpha - \sigma_2)(1 - r)}{pq - (1 - s)(1 - r)}, & \text{if } r \leq 1, \\ \frac{p(\alpha - \sigma_1) + (\alpha - \sigma_2)}{p(q + r) + s - 1}, & \text{if } r > 1 \end{cases}$$

and

$$s_0 := \begin{cases} \frac{q(\alpha - \sigma_2) + (\alpha - \sigma_1)(1 - s)}{pq - (1 - s)(1 - r)}, & \text{if } s \leq 1, \\ \frac{q(\alpha - \sigma_2) + (\alpha - \sigma_1)}{q(p + s) + r - 1}, & \text{if } s > 1. \end{cases}$$

We also define

$$\mathbf{a} := \frac{n}{s_0} \quad \text{and} \quad \mathbf{b} := \frac{n}{r_0}$$

and introduce terminology regarding function pairs (u, v) satisfying (1.1).

- Two solutions u and v to (1.1) are called *decay solutions* whenever there exists $\theta_1, \theta_2 > 0$ such that $u(x) \simeq |x|^{-\theta_1}$ and $v(x) \simeq |x|^{-\theta_2}$.
- (u, v) is said to be *integrable* provided $(u, v) \in L^{\mathbf{a}}(\mathbb{R}^n) \times L^{\mathbf{b}}(\mathbb{R}^n)$.
- (u, v) is called *locally integrable* whenever $(u, v) \in L_{\text{loc}}^{\mathbf{a}}(\mathbb{R}^n) \times L_{\text{loc}}^{\mathbf{b}}(\mathbb{R}^n)$.
- A function pair (u, v) satisfying (1.1) is called *fully non-integrable* if and only if $u \notin L^{\mathbf{a}}(\mathbb{R}^n)$ and $v \notin L^{\mathbf{b}}(\mathbb{R}^n)$.
- We say (u, v) is weakly monotone decreasing if $(u, v) \in \mathbb{W}_m(\mathbb{R}^n) \times \mathbb{W}_m(\mathbb{R}^n)$.

- A classical solution pair is a function pair (u, v) satisfying (1.1) such that $u, v \in C^{[\alpha]}(\mathbb{R}^n)$.

Under our assumptions r_0 and s_0 are well defined positive real numbers. We should also like to point out that when $r = s = 0 \equiv \Gamma_1 \equiv \Gamma_2$ we have $s_0 = q_0$ and $r_0 = p_0$, where p_0 and q_0 are constants defined in [1]. In this same paper it was shown that

$$u(x) \lesssim |x|^{-q_0} \quad \text{and} \quad v(x) \lesssim |x|^{-p_0}$$

whenever u and v are positive, bounded decay solutions to

$$\begin{cases} u(x) = \int_{\mathbb{R}^n} \frac{v(\xi)^q}{|x - \xi|^{n-\alpha} |\xi|^{\sigma_1}} d\xi, & \text{in } \mathbb{R}^n, \\ v(x) = \int_{\mathbb{R}^n} \frac{u(\xi)^p}{|x - \xi|^{n-\alpha} |\xi|^{\sigma_2}} d\xi, & \text{in } \mathbb{R}^n \end{cases} \quad (1.5)$$

under similar conditions. Furthermore, in this same paper, it was shown that positive solutions u and v to (1.5) satisfy

$$\min \{u(x), v(x)\} \gtrsim |x|^{\alpha-n}.$$

In the section that follows, we seek to extend these same estimates to a more general class of functions satisfying the generalized system (1.1).

In the third section we study the sharp-asymptotic properties of weakly monotone decreasing solutions (u, v) to (1.1) and show that fully non-integrable, but locally integrable, solutions cannot decay faster than $|x|^{-s_0}$ and $|x|^{-r_0}$, respectively. In the spirit of [1, THM-4], we shall prove that all positive, locally integrable, and decaying solutions (u, v) decay with the slow rates s_0 and r_0 when (u, v) is fully non-integrable. Consequently, when $r_0 \neq s_0$, it will be shown that there are no fully non-integrable decay solutions $(u, v) \in L_{\text{loc}}^a(\mathbb{R}^n) \times L_{\text{loc}}^b(\mathbb{R}^n)$ satisfying $u(x) \simeq v(x)$.

Subsequently, in the natural case where $\Gamma_1(x, u, v) = \sum_{j=0}^{n_1} a_j(x) u^{r_j}(x) v^{q_j}(x)$ and $\Gamma_2(x, u, v) = \sum_{j=0}^{n_2} b_j(x) u^{r_j}(x) v^{q_j}(x)$, we give conditions for which there cannot exist weakly monotone decreasing function pairs solving (1.1). When $0 \leq r, s \leq 1$, we show that this system admits no $\mathbb{W}_m(\mathbb{R}^n)$ solutions if $0 \leq pq \leq (1-r)(1-s)$. In the special case $pq = 0$, we prove that no positive solutions exist. Finally, using these aforementioned conditions, we partially extend these non-existence results to integrable decay solutions in particular cases of (1.1).

2 Weakly Monotone Decreasing Solution Pairs

In this section we estimate the decay rates of (positive) weakly monotone decreasing functions (u, v) solving system (1.1). We would also like to point out that the results presented in this section are by no means “new”. In fact, almost all of the results presented in this section can be found in [1, §2] or [5] (where it is assumed that $r = s = 0$, $\phi, \psi \equiv 1$ and $\Gamma_{1,2} \equiv 0$), under stronger assumptions. For the remainder of this paper, we shall assume that our positive solution pairs to (1.1) satisfy the following assumption.

Assumption 1. The complement of $\{u < \infty\} \cap \{v < \infty\}$ is a bounded set with Lebesgue measure zero.

Remark 2. Clearly, we require that u and v be finite almost everywhere to avoid the trivial solutions $u \equiv v \equiv \infty$. To see why $\{u < \infty\}^c \cup \{v < \infty\}^c$ should be bounded, we appeal to the weakly decreasing property of u and v . Suppose $f \in W_m(\mathbb{R}^n)$ takes infinite values on an unbounded set. Then there exists a sequence $\{x_k\}_{k=1}^\infty$, tending to infinity, such that $f(x_k) = \infty$ for each $k \geq 1$. Let $R > 0$ and $C > 0$ be such that $f(x) \leq Cf(y)$ whenever $|x| \geq |y| \geq R$. Given $|y| \geq R$ choose an index k such that $|x_k| \geq |y|$ and observe that $f(x_k) \leq Cf(y)$ whence $f(y) = \infty$. This would imply that $f = \infty$ on a set of positive measure. Thus, if one of u or v achieves infinite values on an unbounded set u and v are the trivial infinite solutions.

We note that this assumption is weaker than that made in Villavert [1] throughout the majority of the paper; there it was assumed that u and v were bounded decay functions. Ultimately, it is our intention to establish the following generalization of [1, THM-1]:

Theorem I. *Let (u, v) be a positive solution pair to (1.1). There holds*

$$\begin{aligned} u(x) &\gtrsim (1 + |x|)^{-\min\{n-\alpha, (q+r)(n-\alpha)-(\alpha-\sigma_1)\}}, \\ v(x) &\gtrsim (1 + |x|)^{-\min\{n-\alpha, (p+s)(n-\alpha)-(\alpha-\sigma_2)\}}. \end{aligned}$$

Suppose, in addition, that u and v are weakly monotone decreasing with $p, q > 0$.

- (1) *If $s \leq 1$ we have $u(x) \lesssim |x|^{-s_0}$ and if $r \leq 1$ there holds $v(x) \lesssim |x|^{-r_0}$.*
- (2) *If $u(x) \lesssim v(x)$ we have $u(x) \lesssim |x|^{-s_0}$. Similarly, we have $v(x) \lesssim |x|^{-r_0}$ if $v(x) \lesssim u(x)$.²*

2. Note that the condition in this statement applies, in particular, to decay solutions. Indeed, if u and v are decay solutions we necessarily have either $u \lesssim v$ or $v \lesssim u$.

Surprisingly, it is possible to establish this result by closely mimicking the beautiful arguments used by Villavert in the second section of [1] with some minor tweaking since we do not know a priori how our solutions u and v behave at infinity. The first part of this theorem requires a stronger form of [1, PROP-1].

Proposition 1. *Let (u, v) be a weakly monotone decreasing solution pair to system (1.1) with $p, q > 0$ and $r, s \in [0, 1]$. Then $u(x) \lesssim |x|^{-s_0}$ and $v(x) \lesssim |x|^{-r_0}$.*

Proof. We closely follow the strategy illustrated by Villavert. Observing that u and v satisfy the weak monotone property by hypothesis we are free to choose R and C positive such that u and v satisfy

$$Cu(x) \leq u(y) \quad \text{and} \quad Cv(x) \leq v(y)$$

whenever $|x| \geq |y| \geq R$. Moreover, we are free to assume that u and v are finite valued for all $|x| \geq R$ and that

$$\min \{\phi(x), \psi(x)\} \geq \gamma_0 > 0, \quad \forall |x| \geq R.$$

For $|x| \geq 2R$ we define an annulus in space

$$A_x := \{y \in \mathbb{R}^n : R < |y| < |x|\}$$

whence

$$u(x) \geq \gamma_0 \int_{A_x} \frac{v(y)^q u(y)^p}{|x-y|^{n-\alpha} |y|^{\sigma_1}} dy.$$

Now, using the weakly monotone decreasing property we find

$$u(x) \geq \int_{A_x} \frac{Cv(x)^q u(x)^p}{|x-y|^{n-\alpha} |y|^{\sigma_1}} dy \geq Cu(x)^r v(x)^q |x|^{\alpha-n-\sigma_1} \text{meas}(A_x),$$

where we have used $|x-y| \leq 2|x|$ and $|y| \leq |x|$. Since $|x| \sim |x| - R$ we also discover

$$u(x) \geq Cu(x)^r v(x)^q |x|^{\alpha-\sigma_1}, \quad \text{as } |x| \rightarrow \infty. \quad (2.1)$$

By symmetry of the system, a similar argument grants us the following on v

$$v(x) \geq Cu(x)^p v(x)^s |x|^{\alpha-\sigma_2}, \quad \text{as } |x| \rightarrow \infty. \quad (2.2)$$

CASE 1: $r, s \in [0, 1]$. Using (2.1) and (2.2) we have, as $|x| \rightarrow \infty$,

$$u(x) \geq Cv(x)^{\frac{q}{1-r}} |x|^{\frac{\alpha-\sigma_1}{1-r}} \quad \text{and} \quad v(x) \geq Cu(x)^{\frac{p}{1-s}} |x|^{\frac{\alpha-\sigma_2}{1-s}}.$$

Combining these inequalities yields (for such $|x|$ large)

$$u(x) \geq C u(x)^{\frac{pq}{(1-s)(1-r)}} |x|^{\frac{q(\alpha-\sigma_2)}{(1-s)(1-r)} + \frac{\alpha-\sigma_1}{1-r}}.$$

The above implies that as $|x| \rightarrow \infty$ there holds

$$u(x)^{\frac{pq-(1-s)(1-r)}{(1-s)(1-r)}} \leq C |x|^{-\frac{q(\alpha-\sigma_2)+(\alpha-\sigma_1)(1-s)}{(1-s)(1-r)}}.$$

Whence, as $|x| \rightarrow \infty$

$$u(x) \leq C |x|^{-\frac{q(\alpha-\sigma_2)+(\alpha-\sigma_1)(1-s)}{pq-(1-s)(1-r)}} = C |x|^{-s_0}.$$

Moreover, a symmetric argument shows that $v(x) \lesssim |x|^{-r_0}$ in this case.

CASE 2: $r = 1$ or $s = 1$. Without loss of generality assume $r = 1$ and invoke equation (2.1) to find that, after a correction of C ,

$$v(x) \leq C |x|^{-\frac{\alpha-\sigma_1}{q}} = C |x|^{-r_0}, \quad \text{as } |x| \rightarrow \infty. \quad (2.3)$$

Similarly, if $s = 1$ we use (2.2) and take roots to obtain

$$u(x) \leq C |x|^{-\frac{\alpha-\sigma_2}{p}} = C |x|^{-s_0}, \quad \text{as } |x| \rightarrow \infty.$$

On the other hand, if $0 \leq s < 1$, it follows from (2.3) that for all suitably large x

$$v(x)^{1-s} \leq C |x|^{-\frac{(\alpha-\sigma_1)(1-s)}{q}}.$$

Combining the above estimate with (2.2) grants us the following for all $|x|$ large

$$C u(x)^p |x|^{\alpha-\sigma_2} \leq v(x)^{1-s} \leq C' |x|^{-\frac{(\alpha-\sigma_1)(1-s)}{q}}$$

whence we have

$$u(x)^p \leq C |x|^{-\frac{q(\alpha-\sigma_2)+(\alpha-\sigma_1)(1-s)}{q}}, \quad \text{as } |x| \rightarrow \infty.$$

Taking roots we obtain

$$u(x) \leq C |x|^{-\frac{q(\alpha-\sigma_2)+(\alpha-\sigma_1)(1-s)}{pq}} = C |x|^{-s_0}, \quad \text{as } |x| \rightarrow \infty.$$

A verbatim argument applies to the case of $s = 1$ and $0 \leq r < 1$. This completes the proof. $\square$

An immediate consequence of the above proposition is the subsequent corollary.

Corollary 1. *Let u and v be weakly monotone decreasing functions satisfying system (1.1) when $p, q > 0$ and $r, s \in [0, 1]$. Then*

$$\lim_{|x| \rightarrow \infty} u(x) = \lim_{|x| \rightarrow \infty} v(x) = 0.$$

Remark 3. We would like to point out that in [1] a weaker version of the previous proposition (only the case $r = s = 0$) was established under decay assumptions. In particular, it was assumed that u and v vanish at infinity; in our case no such assumption (in fact, we made no presumptions on the decay rates) was made.

Remark 4. Assume (u, v) is a weakly monotone decreasing solution pair to (1.1). It is not too difficult to show that either $\lim_{|x| \rightarrow \infty} u(x) = 0$ or $\lim_{|x| \rightarrow \infty} v(x) = 0$. Indeed, if

$$\liminf_{|x| \rightarrow \infty} u(x) > 0 \quad \text{and} \quad \liminf_{|x| \rightarrow \infty} v(x) > 0$$

then by taking $R > 0$ large enough and $|x| < R$ we would find

$$\begin{aligned} u(x) &\geq C \int_{|y| \geq R} \frac{u(y)^r v(y)^q}{|x - y|^{n-\alpha} |y|^{\sigma_1}} dy \geq C \int_{|y| \geq R} \frac{1}{|y|^{n-\alpha+\sigma_1}} dy \\ &= C \int_R^\infty \rho^{\alpha-\sigma_1-1} d\rho = \infty. \end{aligned}$$

Hence, either $\liminf_{|x| \rightarrow \infty} u(x) = 0$ or $\liminf_{|x| \rightarrow \infty} v(x) = 0$. We now claim that $f(x) \rightarrow 0$ as $|x| \rightarrow \infty$ whenever $f \in \mathbb{W}_m(\mathbb{R}^n)$ and $\liminf_{|x| \rightarrow \infty} f(x) = 0$. To see that this is so, let $\varepsilon > 0$ be given and pick $R, C > 0$ so that $f(x) \leq Cf(y)$ whenever $|x| \geq |y| \geq R$. Choose y with $|y| \geq R$ such that

$$f(y) < \frac{\varepsilon}{C}.$$

Therefore, whenever $|x| \geq |y|$ we have $f(x) \leq Cf(y) \leq \varepsilon$.

Proposition 2. *Assume $(u, v) \in \mathbb{W}_m(\mathbb{R}^n) \times \mathbb{W}_m(\mathbb{R}^n)$. If $s \in [0, 1]$ and $r > 1$ we have $u(x) \lesssim |x|^{-s_0}$. Similarly, if $r \in [0, 1]$ and $s > 1$ there holds $v(x) \lesssim |x|^{-r_0}$.*

Proof. By symmetry, we may suppose that $s \in [0, 1]$ and $r > 1$ (with $p, q \geq 0$). The argument from the proof of Proposition 1 gives

$$u(x) \geq Cu(x)^r v(x)^q |x|^{\alpha-\sigma_1}, \quad \text{as } |x| \rightarrow \infty.$$

Or, rather,

$$u(x) \leq C v(x)^{\frac{q}{1-r}} |x|^{\frac{\alpha-\sigma_1}{1-r}}, \quad \text{as } |x| \rightarrow \infty. \quad (2.4)$$

and

$$v(x) \geq C u(x)^p v(x)^s |x|^{\alpha-\sigma_2}, \quad \text{as } |x| \rightarrow \infty. \quad (2.5)$$

We now handle two distinct cases.

CASE 1: $s < 1$. In this case we have, as $|x| \rightarrow \infty$,

$$v(x) \geq C u(x)^{\frac{p}{1-s}} |x|^{\frac{\alpha-\sigma_2}{1-s}}$$

whence

$$v(x)^{\frac{q}{1-r}} \leq C u(x)^{\frac{pq}{(1-s)(1-r)}} |x|^{\frac{q(\alpha-\sigma_2)}{(1-s)(1-r)}}, \quad \text{as } |x| \rightarrow \infty.$$

Putting this together with (2.4), we obtain for all x large in norm

$$u(x) \leq C u(x)^{\frac{pq}{(1-s)(1-r)}} |x|^{\frac{q(\alpha-\sigma_2)}{(1-s)(1-r)} + \frac{\alpha-\sigma_1}{1-r}}$$

thereby implying that as $|x| \rightarrow \infty$

$$u(x)^{\frac{(1-s)(1-r)-pq}{(1-s)(1-r)}} \leq C |x|^{\frac{q(\alpha-\sigma_2)+(1-s)(\alpha-\sigma_1)}{(1-s)(1-r)}}.$$

Noting that $(1-s)(1-r) < 0$ and $pq \geq 0$, we obtain that

$$u(x) \leq C |x|^{-\frac{q(\alpha-\sigma_2)+(1-s)(\alpha-\sigma_1)}{pq-(1-s)(1-r)}} = C |x|^{-s_0}.$$

CASE 2: $s = 1$. Note that (2.5) gives

$$u(x) \leq C |x|^{-\frac{\alpha-\sigma_2}{p}} = C |x|^{-s_0}, \quad \text{as } |x| \rightarrow \infty.$$

One can make use of a symmetric argument if $r \in [0, 1]$ and $s > 1$. This concludes the proof. $\square$

We note that these last two propositions when combined prove Theorem I-(1).

Proposition 3. *Let u and v be weakly monotone decreasing functions satisfying (1.1). If $u(x) \lesssim v(x)$ then $u(x) \lesssim |x|^{-s_0}$. If $v(x) \lesssim u(x)$ we have $v(x) \lesssim |x|^{-r_0}$.*

Proof. Without loss of generality assume $u(x) \lesssim v(x)$ and $s > 1$. By using arguments from the previous propositions, we find again that for some constant $C > 0$

$$u(x) \geq Cu(x)^r v(x)^q |x|^{\alpha-\sigma_1} \quad \text{and} \quad v(x) \geq u(x)^p v(x)^s |x|^{\alpha-\sigma_2}, \quad \text{as } |x| \rightarrow \infty.$$

Since $v(x) \gtrsim u(x)$, the second equation gives us

$$v(x) \geq Cu(x)^{p+s} |x|^{\alpha-\sigma_2}, \quad \text{as } |x| \rightarrow \infty.$$

Inserting this into our first equation grants us, as $|x| \rightarrow \infty$,

$$u(x) \geq Cu^r \left(u(x)^{p+s} |x|^{\alpha-\sigma_2} \right)^q |x|^{\alpha-\sigma_1} = Cu^{q(p+s)+r} |x|^{q(\alpha-\sigma_2)+(\alpha-\sigma_1)}.$$

Thus,

$$u(x)^{1-q(p+s)-r} \gtrsim |x|^{q(\alpha-\sigma_2)+(\alpha-\sigma_1)}$$

which gives us the desired inequality for u . A similar argument applies when $v(x) \lesssim u(x)$. $\square$

Before the proof of our first theorem we give a final lemma, whose statement is near verbatim to Proposition 2 of [1] and the proof goes in its spirit, with some additional ideas. Loosely speaking, this lemma gives an upper-bound on the decay rates of u and v .

Lemma 1. *Let (u, v) be a positive solution pair to (1.1). There holds*

$$\min \{u(x), v(x)\} \gtrsim \frac{1}{(1 + |x|)^{n-\alpha}}.$$

Proof. We may take $R > 0$ such that

$$\min \{\phi(x), \psi(x)\} \geq \gamma_0 > 0$$

whenever $|x| \geq R - 1$. Once again, we define an annulus in $\mathbb{R}^n$

$$A := \{y \in \mathbb{R}^n : R - 1 < |y| < R\}.$$

Put $\delta := R$ and note that $|x - y| \leq |x| + \delta$, whence

$$u(x) \geq \gamma_0 \int_A \frac{v(y)^q u(y)^r}{|x - y|^{n-\alpha} |y|^{\sigma_1}} dy \geq \frac{C}{(\delta + |x|)^{n-\alpha}} \int_A \frac{v(y)^q u(y)^r}{|y|^{\sigma_1}} dy.$$

Note that $\int_A \frac{v(y)^q u(y)^r}{|y|^{\sigma_1}} dy$ is a finite constant independent of x , thereby yielding the desired inequality for u . By a symmetric argument, the same inequality holds true for v . $\square$

We are now capable of proving our generalized version of [1, THM-1].

Proof of Theorem I. We begin by establishing the lower bounds on positive solution pairs. For $|x| > 0$ we define an open ball

$$B_x := \left\{ y \in \mathbb{R}^n : |x - y| < \frac{|x|}{2} \right\},$$

and observe that by letting $|x| \rightarrow \infty$ we can make $y \in B_x$ arbitrarily large. Thus, by virtue of Lemma 1, as $|x| \rightarrow \infty$ we have (letting γ_0 be the same as in the previous lemma)

$$\begin{aligned} u(x) &\geq \gamma_0 \int_{B_x} \frac{v(y)^q u(y)^r}{|x - y|^{n-\alpha} |y|^{\sigma_1}} dy \geq \frac{C}{(\delta + |x|)^{(q+r)(n-\alpha)}} \int_{B_x} \frac{1}{|x - y|^{n-\alpha} |y|^{\sigma_1}} dy \\ &\geq \frac{C}{(\delta + |x|)^{(q+r)(n-\alpha) + \sigma_1}} \int_{B_x} \frac{1}{|x - y|^{n-\alpha}} dy. \end{aligned}$$

The estimate for u follows once we observe that

$$\begin{aligned} \int_{B_x} \frac{1}{|x - y|^{n-\alpha}} dy &= \tilde{C} \int_0^{|x|/2} \frac{1}{\rho^{n-\alpha}} \cdot \rho^{n-1} d\rho = \tilde{C} \int_0^{|x|/2} \rho^{\alpha-1} d\rho \\ &= \tilde{C} |x|^\alpha \\ &\sim \tilde{C} (\delta + |x|)^\alpha. \end{aligned}$$

A symmetric argument applies to v .

Assume moreover that u and v are weakly monotone decreasing functions. It is clear that (1) follows from propositions 1 and 2. Finally, (2) follows from Proposition 3. $\square$

3 Sharp Decay Estimates Of $W_m(\mathbb{R}^n)$ Solutions

Throughout this section we suppose that u and v are positive and finite valued outside a bounded set of Lebesgue measure zero, i.e. that u and v satisfy Assumption 1. We also define a condition

$$\begin{cases} \min \{r, s\} \leq 1, \\ u(x) \lesssim v(x), & \text{if } s > 1, \\ v(x) \lesssim u(x), & \text{if } r > 1. \end{cases} \quad (\text{C})$$

Hence, whenever (C) holds true, we have decay estimates on both u and v . Once again, we wish to generalize some of the sharp estimates from §2 of [1] using

some ideas from [3]. More precisely, as in the aforementioned papers, we seek to characterize the decay rates of all non-integrable decay solutions and give strict lower-bounds on the decay rates of $W_m(\mathbb{R}^n)$ -class solution pairs. We would also like to point out that much of the proofs in this section follow the general path outlined by Villavert.

We begin this section with the following lemma, which is a simple calculation taken from [1].

Lemma 2. *Let $(u, v) \in L_{loc}^a(\mathbb{R}^n) \times L_{loc}^b(\mathbb{R}^n)$ be a fully non-integrable solution pair to (1.1) when $p, q > 0$. Assume that condition (C) holds true. For any $\theta > s_0$ and $\vartheta > r_0$*

$$u(x) \not\lesssim \frac{1}{(1 + |x|)^\theta} \quad \text{and} \quad v(x) \not\lesssim \frac{1}{(1 + |x|)^\vartheta}.$$

Proof. As mentioned above, we borrow the proof from [1, PROP-9] and argue by contradiction. We may assume without loss of generality that $u(x) \lesssim |x|^{-\theta}$ for some $\theta > s_0$. Given $R > 0$ sufficiently large compute directly

$$\|u\|_{L^a}^a = \int_{|x| \leq R} u(x)^a dx + \int_{|x| > R} u(x)^a dx = C_1(R) + \int_{|x| > R} u(x)^a dx$$

where $0 \leq C_1(R) < \infty$. In this case, by letting R be sufficiently large, we would find that

$$\begin{aligned} \|u\|_{L^a}^a &\leq C_1(R) + \int_{|x| > R} \frac{C}{(1 + |x|)^{a\theta}} dx \leq C_1(R) + \int_{|x| > R} \frac{C}{|x|^{a\theta}} dx \\ &= C_1(R) + C_2 \int_R^\infty \frac{1}{\rho^{a\theta}} \cdot \rho^{n-1} d\rho \\ &= C_1(R) + C_2 \int_R^\infty \rho^{-\theta a + n - 1} d\rho. \end{aligned}$$

Now, $\theta a > n$ so that $-\theta a - n < 0$ and $\|u\|_{L^a} < \infty$, contradicting $u \notin L^a(\mathbb{R}^n)$. One can make use of a symmetric argument to prove the claim for v ; the proof is now complete. $\square$

We are now prepared to present our generalization of Theorem 4 in [1]: the main result of this section. Essentially, the following theorem sharpens the estimates derived previously and characterizes the decay rates of fully non-integrable (but locally integrable) decay solutions. The statement is nearly identical to the elegant result given by Villavert in [1] and Lei-Li in [5].

Theorem II. *Assume that $p, q > 0$ and assume condition (C) holds true.*

- (1) Suppose (u, v) is a weakly monotone decreasing solution pair to (1.1). For each $\theta_1 \in (0, s_0)$ and $\theta_2 \in (0, r_0)$ there cannot exist $c > 0$ such that

$$u(x) \geq c|x|^{-\theta_1} \quad \text{or} \quad v(x) \geq c|x|^{-\theta_2}, \quad \text{as } |x| \rightarrow \infty.$$

- (2) Let $(u, v) \in L_{loc}^a(\mathbb{R}^n) \times L_{loc}^b(\mathbb{R}^n)$ be a fully non-integrable decay solution pair to (1.1). Then u and v have the slow rates, i.e.

$$u(x) \simeq |x|^{-s_0} \quad \text{and} \quad v(x) \simeq |x|^{-r_0}.$$

A consequence is the following corollary:

Corollary 2. Let $(u, v) \in L_{loc}^a(\mathbb{R}^n) \times L_{loc}^b(\mathbb{R}^n)$ be a fully non-integrable solution pair to (1.1) when $r_0 \neq s_0$. If $u(x) \simeq v(x)$, neither u nor v is a decay solution.

Proof. In this case, note that u is a decay solution if and only if v is as well. By grace of Theorem II we have $u(x) \simeq |x|^{-s_0}$ and $v(x) \simeq |x|^{-r_0}$. Since ‘ $\simeq$ ’ is an equivalence relation, it follows that $|x|^{-s_0} \simeq |x|^{-r_0}$. This is absurd since $r_0 \neq s_0$. $\square$

We shall now give the proof of our second theorem.

Proof of Theorem II. There are two parts to give. We begin by proving the first:

- (1) Suppose, by way of contradiction, that $u(x) \geq c|x|^{-\theta_1}$ as $|x| \rightarrow \infty$ for some $c > 0$ and $\theta_1 \in (0, s_0)$. Recall that by Theorem I we may choose $C > 0$ such that $u(x) \leq C|x|^{-s_0}$ for all sufficiently large x . Combining these inequalities yields:

$$c|x|^{-\theta_1} \leq u(x) \leq C|x|^{-s_0} \quad (\text{as } |x| \rightarrow \infty).$$

Thus, $0 \leq C|x|^{-s_0} - c|x|^{-\theta_1}$ as $|x| \rightarrow \infty$. On the other hand, since $\theta_1 < s_0$, it is clear that

$$c|x|^{-\theta_1} > C|x|^{-s_0}$$

for all large x , thus contradicting our first inequality. A verbatim argument gives us the inequality in the place of v .

- (2) Suppose that $u(x) \simeq |x|^{-\theta}$ for $\theta > 0$. By part (a) we clearly require $\theta \geq s_0$. Finally, by Lemma 2 we cannot have $\theta > s_0$ whence $\theta = s_0$. A symmetric argument grants us $v(x) \simeq |x|^{-r_0}$, as desired. $\square$

4 Non-Existence of Weakly Monotone Decreasing Solution Pairs

Throughout this section it is assumed that u and v satisfy Assumption 1.

In this section we give sufficient conditions on system (1.1) for the non-existence of weakly monotone decreasing super-solutions u and v . For the most part, we shall make additional assumptions on Γ_1 and Γ_2 . More precisely, we are interesting in weakly monotone decreasing functions that satisfy

$$\begin{cases} u(x) \geq \int_{\mathbb{R}^n} \frac{a_1(\xi)v(\xi)^{q_1}u(\xi)^{r_1} + \dots + a_{n_1}(\xi)v(\xi)^{q_{n_1}}u(\xi)^{r_{n_1}}}{|x - \xi|^{n-\alpha} |\xi|^{\sigma_1}} d\xi, & \text{in } \mathbb{R}^n, \\ v(x) \geq \int_{\mathbb{R}^n} \frac{b_1(\xi)u(\xi)^{p_1}v(\xi)^{s_1} + \dots + b_{n_2}(\xi)u(\xi)^{p_{n_2}}v(\xi)^{s_{n_2}}}{|x - \xi|^{n-\alpha} |\xi|^{\sigma_2}} d\xi, & \text{in } \mathbb{R}^n \end{cases} \quad (4.1)$$

where a_j, b_j are non-negative functions for which there holds

$$\liminf_{|x| \rightarrow \infty} a_j(x) > 0 \quad \text{and} \quad \liminf_{|x| \rightarrow \infty} b_j(x) > 0$$

with $p_j, q_j \geq 0$, $r_j, s_j \in [0, 1]$ and $p_j q_j > (1 - s_j)(1 - q_j)$. Systems of a similar type are considered by Quittner in [14] and Montaru-Sirakov-Souplet in [15] for both bounded and infinite domains. As in the previous sections, we also assume $0 < \alpha < n$ and $0 \leq \sigma_1, \sigma_2 < \alpha$. Put for $j \leq \min\{n_1, n_2\}$

$$r_0^{(j)} := \frac{p_j(\alpha - \sigma_1) + (\alpha - \sigma_2)(1 - r_j)}{p_j q_j - (1 - s_j)(1 - r_j)}, \quad s_0^{(j)} := \frac{q_j(\alpha - \sigma_2) + (\alpha - \sigma_1)(1 - s)}{p q - (1 - s_j)(1 - r_j)}.$$

By Theorem I we have for all such j

$$u(x) \lesssim |x|^{-s_0^{(j)}} \quad \text{and} \quad v(x) \lesssim |x|^{-r_0^{(j)}}$$

as well as

$$\begin{aligned} u(x) &\gtrsim \frac{1}{(1 + |x|)^{\min\{n-\alpha, (q_j+r_j)(n-\alpha)-(\alpha-\sigma_1)\}}}, \\ v(x) &\gtrsim \frac{1}{(1 + |x|)^{\min\{n-\alpha, (p_j+s_j)(n-\alpha)-(\alpha-\sigma_2)\}}}, \end{aligned}$$

whenever u and v are weakly monotone decreasing functions. The purpose of this section is to give conditions guaranteeing the non-existence of positive

and/or weakly monotone decreasing solutions to (1.1) and (4.1). These arguments rely strongly on the estimates derived in §2 and extend non-existence results from [11] and [14]. More precisely, we seek to establish a generalization of the strong non-existence result due to Villavert from [11, THM-6]. Originally, this theorem only applied to the case of $\Gamma_{1,2} \equiv 0$, $\phi \equiv \psi \equiv 1$, and $r = s = 0$. However, under additional assumptions, it may be extended to the general system (1.1).

Theorem III.

- (1) Suppose $r, s \in [0, 1]$ and omit the assumption that $pq > (1 - s)(1 - r)$. System (1.1) admits no weakly monotone decreasing solutions when $pq \leq (1 - s)(1 - r)$.
- (2) If $q = 0$ and $r \leq 1$ there are no positive solutions to (1.1). Moreover, this continues to hold if $p = 0$ and $s \leq 1$.
- (3) If there are j and ℓ with $1 \leq j, \ell \leq k$ such that one of the following inequalities hold

- (i) $(q_j + r_j)(n - \alpha) - (\alpha - \sigma_1) < s_0^{(\ell)}$
- (ii) or $(p_j + s_j)(n - \alpha) - (\alpha - \sigma_2) < r_0^{(\ell)}$

there cannot exist a weakly monotone decreasing solution pair to (4.1).

Remark 5. The first part of this theorem tells us that if

$$p_i q_j \leq (1 - r_j)(1 - s_i) \quad \text{for some } i \text{ and } j,$$

there are no weakly monotone decreasing solution pairs to (4.1). Furthermore, there are no positive solutions if $p_i q_j = 0$ for some i, j .

Remark 6. Consider the Liouville-Bratu-Gelfand equation

$$\begin{cases} \Delta u + \mu e^{\lambda u} \equiv 0, & \text{in } \mathbb{R}^n, \\ u > 0, & \text{in } \mathbb{R}^n, \\ u \in C^2(\mathbb{R}^n) \end{cases} \quad (4.2)$$

in $n \geq 3$ dimensions with $\mu, \lambda > 0$. This is “equivalent” to $u \in C^2(\mathbb{R}^n)$ being a positive solution to (see [11, THM-1])

$$u(x) = \int_{\mathbb{R}^n} \frac{\mu e^{\lambda u(y)}}{|x - y|^{n-2}} dy \quad \text{for } x \in \mathbb{R}^n.$$

However, since $e^x \geq x$ for all $x \in \mathbb{R}$, this would imply

$$u(x) \geq \int_{\mathbb{R}^n} \frac{\mu \lambda u(y)}{|x - y|^{n-2}} dy, \quad \text{in } \mathbb{R}^n.$$

which is impossible by the second part of Theorem III. Hence, system (4.2) admits no solution.

Furthermore, recalling the inequality $\sinh x \geq x$ for $x \geq 0$, it is clear that we may apply the same argument to deduce that the following sinh-Poisson type system,

$$\begin{cases} \Delta u + \lambda^2 \sinh u \equiv 0, & \text{in } \mathbb{R}^n, \\ u > 0, & \text{in } \mathbb{R}^n, \\ u \in C^2(\mathbb{R}^n), \end{cases}$$

has no solution with $\lambda \neq 0$ and $n \geq 3$.

We establish this theorem as the combination of propositions, for the sake of simplicity. However, we must first provide a useful lemma regarding the asymptotic behaviour of non-trivial weakly monotone decreasing functions.

Lemma 3. *Let $f : \mathbb{R}^n \rightarrow (0, \infty]$ be a weakly monotone decreasing function satisfying Assumption 1. Then*

$$\limsup_{|x| \rightarrow \infty} f(x) < \infty.$$

Proof. Since f is weakly monotone decreasing we may take $y \in \mathbb{R}$ so large in norm that $f(x) \leq C f(y)$ whenever $|x| \geq |y|$, where C is some positive constant independent of x . Without loss of generality suppose that $f(y) < \infty$. This implies that $\limsup_{|x| \rightarrow \infty} f(x) \leq C f(y) < \infty$ as was asserted. $\square$

Remark 7. Actually, it follows that there cannot exist a weakly monotone decreasing solution pair (u, v) to system (1.1) if either

$$(q + r)(n - \alpha) - (\alpha - \sigma_1) \leq 0 \quad \text{or} \quad (p + s)(n - \alpha) - (\alpha - \sigma_2) \leq 0.$$

Indeed, by the second part of Theorem I, we would have $\lim_{|x| \rightarrow \infty} u(x) = \infty$ or $\lim_{|x| \rightarrow \infty} v(x) = \infty$.

Proposition 4. *Assume $r, s \leq 1$. System (1.1) does not admit any non-trivial weakly monotone decreasing solution pairs when $0 < pq \leq (1 - r)(1 - s)$.*

Proof. For this proof we borrow ideas from [1, PROP-8] and [11, THM-6]. Since we are handling the case $pq > 0$ we are assuming, especially, that $r, s \in [0, 1)$. Suppose, by way of contradiction, that $(u, v) \in \mathbb{W}_m(\mathbb{R}^n) \times \mathbb{W}_m(\mathbb{R}^n)$ is a solution pair to system (1.1) when $pq \leq (1-r)(1-s)$. Fix $R > 0$ sufficiently large and let $|x| \geq 2R$, say. One has for suitable $c > 0$

$$u(x) \geq \int_{B_R(0)} \frac{\phi(y)v(y)^q u(y)^r}{|x-y|^{n-\alpha} |y|^{\sigma_1}} dy \geq \frac{c}{|x|^{n-\alpha}} \int_{B_R(0)} \frac{\phi(y)v(y)^q u(y)^r}{|y|^{\sigma_1}} dy = \frac{c}{|x|^{n-\alpha}} = c|x|^{-b_0}.$$

as $|x| \rightarrow \infty$ for $b_0 = n - \alpha$. Now, for x very large we define

$$B_x := \{y \in \mathbb{R}^n : R < |y| < |x|\}$$

and consider

$$\begin{aligned} v(x) &\geq C v(x)^s \int_{B_x} \frac{|y|^{-pb_0-\sigma_2}}{|x-y|^{n-\alpha}} dy \geq C v(x)^s |x|^{-pb_0-\sigma_2} \int_R^{|x|} \rho^{\alpha-1} d\rho \\ &\geq C v(x)^s |x|^{-(pb_0-\alpha+\sigma_2)}, \quad \text{as } |x| \rightarrow \infty. \end{aligned}$$

Hence,

$$v(x) \geq C |x|^{-a_1}, \quad a_1 := \frac{pb_0 - \alpha + \sigma_2}{1-s}$$

as $|x| \rightarrow \infty$. By taking R sufficiently large in each step, one can find by induction that, as $|x| \rightarrow \infty$, one has

$$u(x) \gtrsim |x|^{-b_k} \quad \text{and} \quad v(x) \gtrsim |x|^{-a_k}$$

where

$$a_{k+1} := \frac{pb_k - \alpha + \sigma_2}{1-s} \quad \text{and} \quad b_k := \frac{qa_k - \alpha + \sigma_1}{1-r}.$$

The idea is to rewrite the induced recurrence relation in simpler terms to estimate b_k . Let us now define

$$P := \frac{p}{1-s}, \quad Q := \frac{q}{1-r}, \quad \Sigma_1 := \frac{\sigma_1}{1-r}, \quad \Sigma_2 := \frac{\sigma_2}{1-s}, \quad A := \frac{\alpha}{1-s} \quad \text{and} \quad B := \frac{\alpha}{1-r}.$$

Using the above notation, we rewrite the recurrence relation of interest in the following way

$$a_{k+1} := Pb_k + \Sigma_2 - A, \quad b_k := Qa_k + \Sigma_1 - B.$$

By way of determining a closed form, let $k \in \mathbb{N}$ be large and $1 \leq j \leq k$ an integer. The reader may verify by direct substitution that

$$b_k = Q^j P^j b_{k-j} + \left(Q + Q^2 P + Q^3 P^2 + \cdots + Q^j P^{j-1} \right) (\Sigma_2 - A) \\ + \left(1 + QP + \cdots + Q^{j-1} P^{j-1} \right) (\Sigma_1 - B)$$

Now, taking $j = k$ we find

$$b_k = (PQ)^k b_0 + Q (\Sigma_2 - A) \sum_{\ell=0}^{k-1} (PQ)^\ell + (\Sigma_1 - B) \sum_{\ell=0}^{k-1} (PQ)^\ell. \quad (4.3)$$

Which yields the following simple expression for b_k

$$b_k = (PQ)^k b_0 + [Q (\Sigma_2 - A) + (\Sigma_1 - B)] \sum_{\ell=0}^{k-1} (PQ)^\ell$$

There are now two cases to distinguish.

CASE 1: Assume $pq = (1-s)(1-r)$. Then $PQ = 1$ so that $b_k \rightarrow -\infty$ as $k \rightarrow \infty$.

CASE 2: Suppose $pq < (1-s)(1-r)$. We then have

$$0 < PQ = \frac{pq}{(1-s)(1-r)} < 1$$

whence

$$b_k = (PQ)^k b_0 + [Q(\Sigma_2 - A) + (\Sigma_1 - B)] \frac{(PQ)^k - 1}{PQ - 1}$$

Now, we calculate

$$Q (\Sigma_2 - A) + (\Sigma_1 - B) = \frac{q}{1-r} \left(\frac{\sigma_2 - \alpha}{1-s} \right) + \frac{\sigma_1 - \alpha}{1-r} \\ = \frac{q(\sigma_2 - \alpha) + (\sigma_1 - \alpha)(1-s)}{(1-r)(1-s)}.$$

Finally,

$$PQ - 1 = \frac{pq}{(1-s)(1-r)} - 1 = \frac{pq - (1-s)(1-r)}{(1-s)(1-r)}$$

whence

$$\begin{aligned} (Q(\Sigma_2 - A) + (\Sigma_1 - B)) \frac{1}{PQ - 1} &= \frac{q(\sigma_2 - \alpha) + (\sigma_1 - \alpha)(1 - s)}{pq - (1 - s)(1 - r)} \\ &= -s_0. \end{aligned}$$

Under our conditions we have $-s_0 > 0$ implying that $b_k < 0$ for large enough k .

In either case we may make $b_k < 0$ for all $k \in \mathbb{N}$ sufficiently large. Hence, for suitable k there holds

$$u(x) \gtrsim |x|^{-b_k} \quad \text{where } b_k < 0$$

which implies $\lim_{|x| \rightarrow \infty} u(x) = \infty$. Similarly, since $a_{k+1} \leq b_k$ for all $k \in \mathbb{N}$, it is easy to see that $\lim_{|x| \rightarrow \infty} v(x) = \infty$. However, this contradicts Lemma 3. $\square$

Proposition 5. *If $p = 0$ and $r \leq 1$ there does not exist a positive solution pair to (1.1). Similarly, there are no positive solutions if $q = 0$ and $s \leq 1$.*

Remark 8. In the above proposition we do not assume that $pq > (1 - s)(1 - r)$. In fact, if $pq = 0$ we always have $pq \leq (1 - s)(1 - r)$. Furthermore, we do not require this weak monotonicity we have used frequently.

Proof of Proposition. Suppose, without loss of generality, that $q = 0$ and $r < 1$; a similar argument applies when $p = 0$ and $s < 1$. Fix $R > 0$ large and note that from (1.1) we obtain for, say, $|x| \geq 2R$

$$u(x) \geq \int_{B_R(0)} \frac{\phi(y)u(y)^r}{|x - y|^{n-\alpha} |y|^{\sigma_1}} dy \geq \frac{c}{|x|^{n-\alpha}} \int_{B_R(0)} \frac{\phi(y)u(y)^r}{|y|^{\sigma_1}} dy$$

implying that $u(x) \geq c|x|^{-(n-\alpha)}$ as $|x| \rightarrow \infty$ for some constant $c > 0$. Fix $R > 0$ so large that $u(x) \geq c|x|^{-(n-\alpha)}$ whenever $|x| \geq R$. Given $|x| \geq 2R$ we define as in the proof of Theorem I

$$A_x := \{y \in \mathbb{R}^n : R < |y| < |x|\}$$

so that

$$\begin{aligned} u(x) &\geq \int_{A_x} \frac{\phi(y)u(y)^r}{|x - y|^{n-\alpha} |y|^{\sigma_1}} dy \geq c|x|^{-r(n-\alpha)-\sigma_1} \int_{A_x} \frac{1}{|x - y|^{n-\alpha}} dy \\ &\geq c|x|^{-r(n-\alpha)-\sigma_1+\alpha-n} \text{meas}(A_x) \\ &\sim c|x|^{-r(n-\alpha)+\alpha-\sigma_1}, \quad \text{as } |x| \rightarrow \infty. \end{aligned}$$

Or, rather,

$$u(x) \gtrsim |x|^{-(rb_0 + \sigma_1 - \alpha)}, \quad \text{where } b_0 := n - \alpha.$$

Of course, we may repeat this argument inductively on $k \in \mathbb{N}$ to find that

$$u(x) \gtrsim |x|^{-b_k}, \quad \text{where } b_k := rb_{k-1} + \sigma_1 - \alpha \quad (4.4)$$

for all $k \in \mathbb{N}$. By grace of a geometric sum, it is easy to verify that for each $k \in \mathbb{N}$

$$b_k = \begin{cases} r^k b_0 + (\sigma_1 - \alpha) \cdot \frac{1-r^k}{1-r}, & \text{if } r < 1, \\ b_0 + k(\sigma_1 - \alpha), & \text{if } r = 1. \end{cases}$$

Since $\sigma_1 < \alpha$, by taking $k \rightarrow \infty$, we can make $b_k < 0$ for some $k \in \mathbb{N}$. Fix $R > 0$ large and assume $|x| < R$; there then holds

$$\begin{aligned} u(x) &\geq c \int_{B_R(0)^c} \frac{u(y)^r}{|x-y|^{n-\alpha} |y|^{\sigma_1}} dy \geq c \int_{B_R(0)^c} |y|^{-rb_k + \alpha - n - \sigma_1} dy \\ &\geq c \int_R^\infty \rho^{-rb_k + \alpha - \sigma_1 - 1} d\rho \end{aligned}$$

where this last integral is convergent if and only if $-rb_k + \alpha - \sigma_1 < 0$. Hence, we get that $u(x) = \infty$ in $|x| < R$: a contradiction. $\square$

We now prepared to prove our final theorem.

Proof of Theorem III. The first two parts follow from propositions 4 and 5. As for the last part, suppose without loss of generality that

$$(q_j + r_j)(n - \alpha) - (\alpha - \sigma_1) < s_0^{(\ell)}.$$

By our earlier observations, we have

$$\frac{1}{|x|^{(q_j + r_j)(n - \alpha) - (\alpha - \sigma_1)}} \lesssim u(x) \lesssim |x|^{-s_0^{(\ell)}}$$

which is absurd. $\square$

Therefore, to establish non-existence results it suffices to see under which condition there holds these inequalities. Especially, by applying the third part of Theorem III, we deduce two conditions for the non-existence of functions $(u, v) \in W_m(\mathbb{R}^n) \times W_m(\mathbb{R}^n)$ solving (4.1).

Condition 1. Suppose that $n - \alpha \leq \alpha - \sigma_2$ with $r_j = 0$, for some index j . Let ℓ denote an index where q_ℓ is maximal. If $p_\ell q_\ell - (1 - s_\ell)(1 - r_\ell) < 1$ there does not exist a weakly monotone decreasing solution pair to (4.1). Certainly, we write

$$\begin{aligned} (q_j + r_j)(n - \alpha) - (\alpha - \sigma_1) &\leq q_\ell(n - \alpha) - (\alpha - \sigma_1)(1 - s_\ell) \\ &< \frac{q_\ell(n - \alpha) - (\alpha - \sigma_1)(1 - s_\ell)}{p_\ell q_\ell - (1 - s_\ell)(1 - r_\ell)} \\ &\leq \frac{q_\ell(\alpha - \sigma_2) - (\alpha - \sigma_1)(1 - s_\ell)}{p_\ell q_\ell - (1 - s_\ell)(1 - r_\ell)} \\ &= s_0^{(\ell)}. \end{aligned}$$

Luckily, by these same arguments, we are free to (partially) extend a non-existence result from Villavert [11, THM-2]

Condition 2. If there is an index j so that

$$n - \alpha < \max \{r_0^{(j)}, s_0^{(j)}\}$$

there does not exist a weakly monotone decreasing solution pair to (4.1). To see this, assume $s_0^{(j)} > n - \alpha$ for some index j . We may choose $c, C > 0$ such that

$$c|x|^{-(n-\alpha)} \leq u(x) \leq C|x|^{-s_0^{(j)}}, \quad \text{as } |x| \rightarrow \infty.$$

Whence, $0 \leq C|x|^{-s_0^{(j)}} - c|x|^{-(n-\alpha)}$ for all large x , which is absurd since $s_0^{(j)} > n - \alpha$. Similarly one can show that $r_0^{(j)} > n - \alpha$ is impossible for a weakly decreasing solution pair.

Remark 9. Using the above conditions we are able to give non-existence criteria for super-solutions to the Lane-Emden type systems considered by Lei-Li in [5], i.e.

$$\begin{cases} u(x) \geq \int_{\mathbb{R}^n} \frac{v(y)^q}{|x - y|^{n-\alpha}} dy, & \text{in } \mathbb{R}^n, \\ v(x) \geq \int_{\mathbb{R}^n} \frac{u(y)^p}{|x - y|^{n-\alpha}} dy & \text{in } \mathbb{R}^n. \end{cases} \quad (4.5)$$

Indeed, by Condition 1 it suffices to have $pq < 2$ when $n \leq 2\alpha$.³

Furthermore, we may apply these same non-existence conditions to coupled nonlinear systems of the form:

$$\begin{cases} u(x) = \int_{\mathbb{R}^n} \frac{u(y)^r + v(y)^q}{|x - y|^{n-2}} dy, & \text{in } \mathbb{R}^n, \\ v(x) = \int_{\mathbb{R}^n} \frac{v(y)^s + u(y)^p}{|x - y|^{n-2}} dy & \text{in } \mathbb{R}^n \end{cases} \quad (4.6)$$

3. Alternatively, this non-existence result holds when $pq \leq 2$ and $n < 2\alpha$.

with $p, q > 0$, $r, s \in \mathbb{R}$ and $pq > 1$. Indeed, solutions to (4.6) are super-solutions to (4.5) with $\alpha = 2$. Hence, we may conclude that the above system has no $W_m(\mathbb{R}^n)$ -class solution pairs whenever $pq < 2$ and $3 \leq n \leq 4$. We should also note that this system is related (and often equivalent) to the elliptic system considered by Li-Villavert in [6]:

$$\begin{cases} -\Delta u \equiv u^r + v^q & \text{in } \mathbb{R}^n, \\ -\Delta v \equiv v^s + u^p & \text{in } \mathbb{R}^n. \end{cases}$$

5 Integrable Decay Solutions

In this section we study system (1.1) under the same conditions as §2, but focus on solution pairs (u, v) that are decaying and integrable. More precisely, we assume the following throughout this section:

Assumption 2. Suppose that $0 \leq r, s \leq 1$ and that (u, v) are positive, integrable and decaying solutions to (1.1) that are finite outside a bounded set of Lebesgue measure zero.

By previous remarks, we are supposing in particular that u and v are weakly monotone decreasing and hence we are free to apply the estimates derived previously to these solution pairs. In fact, we seek primarily to extend earlier results to the integrable case. We do this as follows.

Proposition 6. *If one of*

$$(i) \quad q(\alpha - \sigma_2) + (\alpha - \sigma_1) \geq pq(n - \alpha) \text{ or,}$$

$$(ii) \quad p(\alpha - \sigma_1) + (\alpha - \sigma_2) \geq pq(n - \alpha)$$

holds, there does not exist an integrable decay solution pair to (1.1).

Proof. By hypothesis on u there exists $\theta > 0$ and positive constants c, C such that

$$c|x|^{-\theta} \leq u(x) \leq C|x|^{-\theta}$$

for all x large in norm. By virtue of Lemma 1 we cannot have $\theta > (n - \alpha)$ for then,

$$c|x|^{-(n-\alpha)} \leq u(x) \leq C|x|^{-\theta} \quad (\text{as } |x| \rightarrow \infty),$$

which is impossible since $C|x|^{-\theta} < c|x|^{-(n-\alpha)}$ for $|x|$ large. Thus, we must have $\theta \leq (n - \alpha)$. We claim that $\theta \leq (n - \alpha)$ is impossible. To prove this, assume by way of a contradiction that

$$u(x) \geq c|x|^{-\theta}, \quad \text{as } |x| \rightarrow \infty$$

for some $\theta \in (0, n - \alpha]$ and $c > 0$. Then,

$$\begin{aligned} \|u\|_{L^{\mathfrak{a}}(\mathbb{R}^n)}^{\mathfrak{a}} &= \int_{\mathbb{R}^n} u(x)^{\mathfrak{a}} dx \geq \int_{|x| \gg 1} u(x)^{\mathfrak{a}} dx \geq c \int_{|x| \gg 1} |x|^{-\theta \mathfrak{a}} dx \\ &\geq c \int_R^{\infty} \rho^{-\theta \mathfrak{a} + n - 1} d\rho. \end{aligned}$$

for $R > 0$ very large. Since $u \in L^{\mathfrak{a}}(\mathbb{R}^n)$ it suffices to show that $-\theta \mathfrak{a} + n \geq 0$ to derive a contradiction. Moreover, we observe that $-\mathfrak{a}\theta \geq -\mathfrak{a}(n - \alpha)$ whence

$$\|u\|_{L^{\mathfrak{a}}(\mathbb{R}^n)}^{\mathfrak{a}} \geq c \int_R^{\infty} \rho^{-\mathfrak{a}(n-\alpha)+n-1} d\rho,$$

for all large R . Let us now note that

$$\begin{aligned} -\mathfrak{a}(n - \alpha) + n &= \frac{-n(n - \alpha) [pq - (1 - s)(1 - r)]}{q(\alpha - \sigma_2) + (\alpha - \sigma_1)(1 - s)} + n \\ &\geq \frac{-n(n - \alpha)pq}{q(\alpha - \sigma_2) + (\alpha - \sigma_1)(1 - s)} + n \\ &\geq \frac{-n(n - \alpha)pq}{q(\alpha - \sigma_2) + (\alpha - \sigma_1)} + n \\ &= n \left(1 - \frac{pq(n - \alpha)}{q(\alpha - \sigma_2) + (\alpha - \sigma_1)} \right). \end{aligned}$$

We wish to show that $\left(1 - \frac{pq(n-\alpha)}{q(\alpha-\sigma_2)+(\alpha-\sigma_1)}\right) \geq 0$, which is equivalent to establishing

$$q(\alpha - \sigma_2) + (\alpha - \sigma_1) \geq pq(n - \alpha).$$

Hence,

$$\int_R^{\infty} \rho^{-\theta \mathfrak{a} + n - 1} d\rho = \infty$$

contradicting the integrability of u . Similarly, one can show that no decaying integrable solution v can exist when (ii) holds. $\square$

Remark 10. We may apply this to the weighted and coupled Schrödinger system

$$\begin{cases} \int_{\mathbb{R}^n} \frac{u(y)^r v(y)^q}{|x-y|^{n-2} |y|^{\sigma_1}} dy, & \text{in } \mathbb{R}^3, \\ \int_{\mathbb{R}^n} \frac{v(y)^s u(y)^p}{|x-y|^{n-2} |y|^{\sigma_2}} dy, & \text{in } \mathbb{R}^3. \end{cases}$$

Under classical conditions this is equivalent to the system:

$$\begin{cases} -\Delta u = \frac{u(x)^r v(x)^q}{|x|^{\sigma_1}}, & x \in \mathbb{R}^3 \setminus \{0\}, \\ -\Delta v = \frac{u(x)^p v(x)^s}{|x|^{\sigma_1}}, & x \in \mathbb{R}^3 \setminus \{0\}. \end{cases}$$

By the previous proposition, if one of

- (i) $q(2 - \sigma_2) + (2 - \sigma_1) \geq pq$, or
- (ii) $p(2 - \sigma_1) + (2 - \sigma_2) \geq pq$

there can be no integrable decay solution pair. In the unweighted case, this becomes the condition

$$2(q + 1) \geq pq \quad \text{or} \quad 2(p + 1) \geq pq.$$

Acknowledgments

The author would like to thank Dana Berman for fruitful discussions and her time spent proof-reading this paper. The author is especially grateful to this his parents: John Chernysh and Jovette Marois for their endless support. Finally, the author is grateful to Professor Vétois at McGill university for his guidance and support.

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